ALGORITHMS FOR SENSOR-BASED ROBOTICS

This course surveys the development of robotic systems for navigating in an environment from an algorithmic perspective. It will cover basic kinematics, configuration space concepts, motion planning, and localization and mapping. It will describe these concepts in the context of the ROS software system, and will present examples relevant to mobile platforms, manipulation, robotics surgery, and human-machine systems.

Credit Hours: 3
Instructor: Simon Leonard
Lectures: Tuesday, Thursday 12h00-1h15PM (Hodson 210)
E-mail: sleonard at jhu dot edu
Office hours: Monday 4h00-5h00PM Thursday 6-7PM Thursday (Zoom link in syllabus)

Course Assistants office hours (Zoom links in the syllabus)

Resources:

Wyman 170: 12 desktops with Ubuntu 22.04 and ROS2 Humble.

(Primary book) Principles of Robot Motion (Choset et. al.), MIT Press, 2005
Additional online resources will be made available for the assignments.

Other resources

Probabilistic Robotics, Thrun, Burgard, Fox, MIT Press, 2006.
Planning Algorithms, Steve Lavalle, Cambridge University Press, 2006. book

ROS https://docs.ros.org/en/humble/index.html

Course Preparation

This course is intended for advanced undergraduates and first year graduate students. I assume students have a rudimentary understanding of linear algebra, calculus, and are able to program in C++ and Matlab.

Goals and Evaluation

There will be five assignments and two exams. Grading will be 47% on the assignments and 25% for each exam and 3% for demonstrations. All students will need to use real robots by the of the term to earn their final mark.

Assignment due dates:

Assignment 1: Feb 12th 2026
Assignment 2: Feb 26th 2026
Assignment 3: Mar 26th 2026
Assignment 4: Apr 14th 2026
Assignment 5: Apr 27th 2026
Midterm: Mar 5th 2026
Final: TBD

Assignments are due by 23:59 EST on the day it is due. **10% per day of the grade is deducted for late submission. Deadlines are provided way ahead of time so work around them. Procrastinate at your own risk.**

Turn in assignments on Canvas. No email.

Canvas is used for asking questions and posting comments.

Schedule

Week 1: Introduction

Week 2: Rigid body motion and velocity. Review of linear algebra used to describe relative 3D motion and velocity between rigid bodies. Kinematics: Forward/inverse kinematics of kinematics chains and mobile robots and manipulator Jacobian.

Suggested readings:

- Choset, H., Lynch, K.M., *Principles of Robot Motion: Theory, Algorithms, and Implementations* (Chapter 3).
- Murray, R.M., Li, Z., Sastry, S.S., A Mathematical Introduction to Robotic Manipulation (Chapter 2).
- Siciliano. B., Khatib, O., *Springer Handbook of Robotics* (Chapter 1).

Week 3: Hand-eye calibration.

- *F. Park, B. Martin* Robot Sensor Calibration: Solving AX=XB on the Euclidian Group. IEEE Trans. Rob.Aut. 10(5):717-721, October 1994

Week 4: Potential fi elds. Attractive and repulsive potentials. Implementation of sensor constraints as potentials.

Suggested readings:

- Choset, H., Lynch, K.M., *Principles of Robot Motion: Theory, Algorithms, and Implementations* (Chapter 4).
- Mezouar, Y., Chaumette, F., *Path Planning for Robust Image-based Control*, IEEE Tr. on Robotics 2002.

Week 5: Graph-based path planning. Visibility maps, generalized Voronoi diagram, cell decompositions and Minkowski sum.

Suggested readings:

- Choset, H., Lynch, K.M., *Principles of Robot Motion: Theory, Algorithms, and Implementations* (Chapters 5, 6).
- de Berg, M., Cheong, O., van Kreveld, M., Overmars, M., *Computational Geometry: Algorithms and Applications* (Chapters 6, 7, 13, 15).

Week 6: Sample-based path planning. Probabilistic roadmaps, rapidly-exploring random trees.

Suggested readings:

- Choset, H., Lynch, K.M., *Principles of Robot Motion: Theory, Algorithms, and Implementations* (Chapters 7).
- LaValle, S., *Planning Algorithms* (Chapter 5).

Week 7: Review and midterm.

Week 8, 9: Kalman filter, dynamic linear systems. Extended Kalman filter, dynamic non-linear systems.

Suggested readings:

- Choset, H., Lynch, K.M., *Principles of Robot Motion: Theory, Algorithms, and Implementations* (Chapters 8).
- Davison, A.J., Reid, I.D., Molton, N.D., Stasse, O., *MonoSLAM: Real-Time Single Camera SLAM*, PAMI 2007.
- Harris, C., *Geometry from visual motion*. In Active Vision 1993.

Week 10, 11, 12: Bayesian filtering and multiple hyphotheses. Particle fi lter.

Suggested readings:

- Choset, H., Lynch, K.M., *Principles of Robot Motion: Theory, Algorithms, and Implementations* (Chapters 9).
- Thrun, S., Fox, D., Burgard, W., Dellaert, F., *Robust Monte Carlo localization for mobile robots*, Artifi cial Intelligence 2000.

- Doucet, A., de Freitas, N., Murphy, K., Russel, S., *Rao-Blackwellized particle filtering for dynamic Bayesian networks*, Uncertainty in Artificial Intelligence 2000.

Week 13: Review

Regarding plagiarism:

The strength of the university depends on academic and personal integrity. In this course, you must be honest and truthful. Ethical violations include cheating on exams, plagiarism, reuse of assignments, improper use of the Internet and electronic devices, unauthorized collaboration, alteration of graded assignments, forgery and falsification, lying, facilitating academic dishonesty, and unfair competition.

In addition, the specific ethics guidelines for this course are:

1. No teamwork or collaboration on assignments
2. Public or private sharing of code, assignments and exams

Report any violations you witness to the instructor. You may consult the associate dean of student conduct (or designee) by calling the Office of the Dean of Students at 410-516-8208 or via email at integrity@jhu.edu. For more information, see the Homewood Student Affairs site on academic ethics: https://studentaffairs.jhu.edu/policies-guidelines/undergrad-ethics

Read the Graduate Academic Misconduct Policy:

Examples of academic misconduct:

- Use of material produced by another person without acknowledging its source
- Submission of the same or substantially similar work of another person (e.g., an author, a classmate, etc.)
- Intentionally or knowingly aiding another student to commit an academic ethics violation
- Allowing another student to copy from one's own assignment, test, or examination
- Making available copies of course materials whose circulation is prohibited (e.g., old assignments, texts or examinations, etc.)

Teamwork for assignments is **not** allowed. You cannot use work from someone else (plagiarism). You cannot share your assignments to someone else (facilitating academic dishonesty).

- You cannot use **any** material from a classmate or student who took the course in previous semesters.
- You cannot share **any** material from this course in future semester.
- You cannot use any online tools (AI-based or otherwise) such as ChatGTP or Copilot.

Any academic misconducts (big or small) will be reported to Graduate Affairs (or Office of the Dean) and will be at least sanctioned by an "F" course grade. No negotiation. I will retroactively sanction plagiarism (i.e. a student plagiarize your assignment in future semesters).

Computational Interaction and Robotics Laboratory.

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